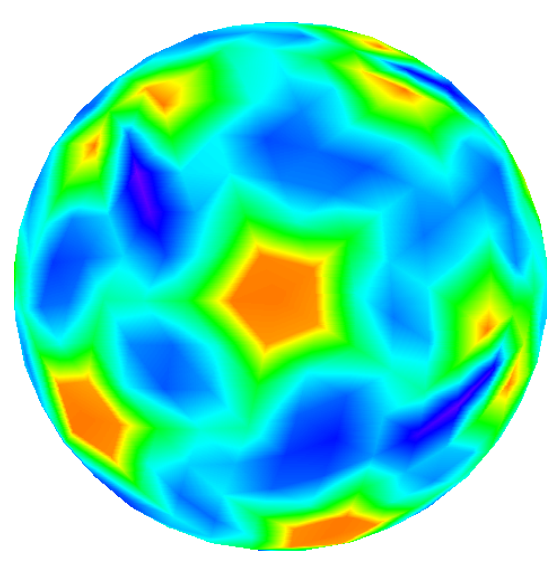


MODELING THE THOMSON PROBLEM WITH CIRCLE PACKINGS



DANIEL VILLANUEVA MERSCHIA GRIFFIN JAMES ASHE, PH.D.
DEPARTMENT OF NATURAL SCIENCES AND MATHEMATICS, JOHNSON C. SMITH UNIVERSITY, CHARLOTTE, NC

THE THOMSON PROBLEM

The Thomson problem asks how to find the minimum energy distribution of N electrons on the surface of the sphere. The *energy* of N points, $p_1, p_2, \dots, p_N$, on the sphere is defined as

$$E = \sum_{i < j} \frac{1}{|p_i - p_j|}.$$

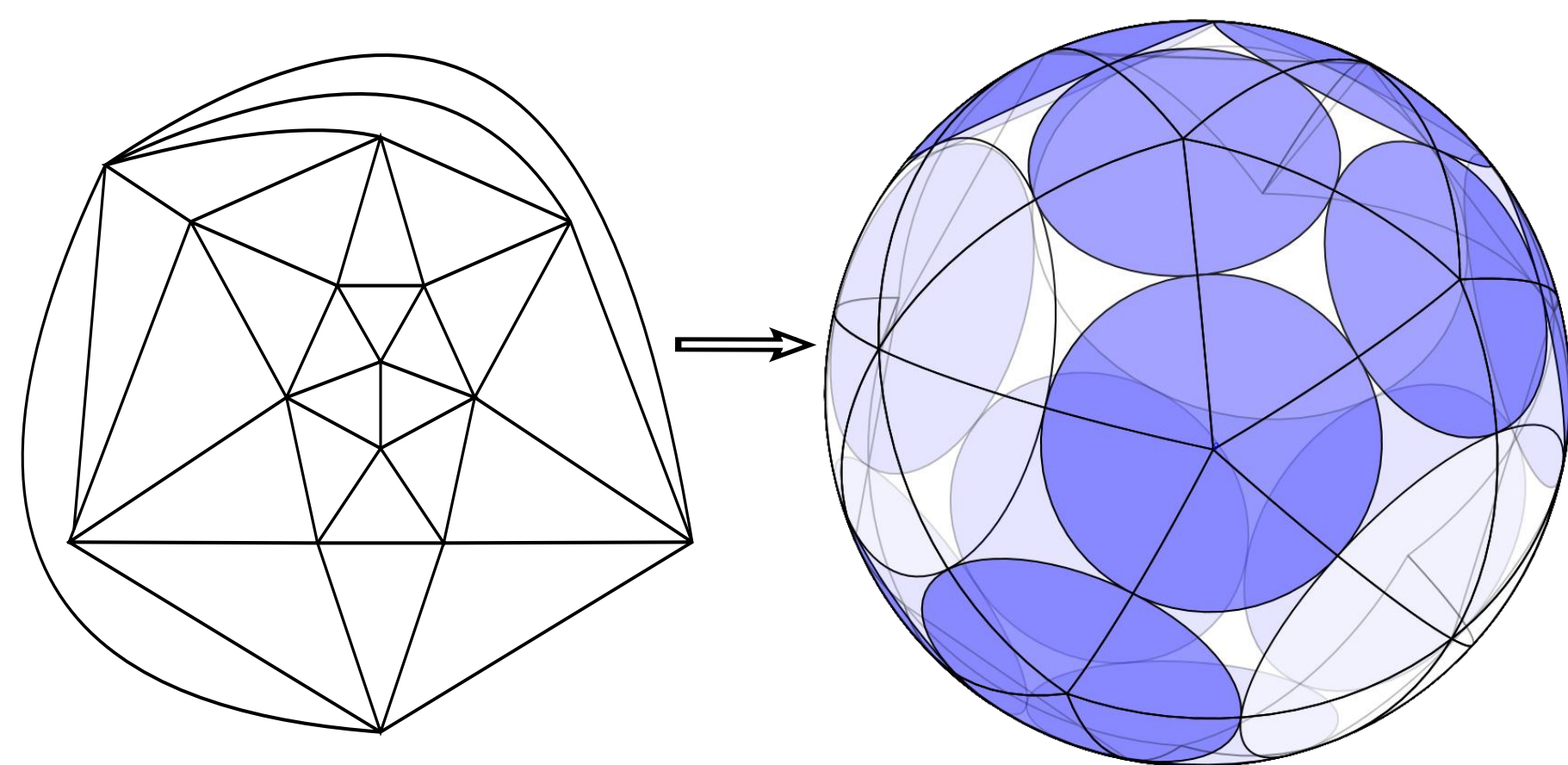
Number 7 on Smale's list of conjectures, this problem remains one of the most important unsolved packing problems, and has applications in biology and physics [1].

Known solutions have been rigorously proven for only a small number of special cases. In general, finding a solution is an NP hard problem, and progress has relied on computational methods [2].

CIRCLE PACKINGS

Circle packings are configurations of circles with specified patterns of tangency. The following theorem provides rigidity and existence of circle packings on the sphere.

Koebe-Andreiev-Thurston Theorem. *There exists a circle packing on the sphere for any finite triangulation of the sphere. Furthermore, this circle packing is unique up to Möbius transformations.*



A tangency pattern and its circle packing.

The actual radii for the circle packing guaranteed by this theorem can always be found using the *Circle Packing Algorithm* [3]. Moreover we have a feature rich software package, *CirclePack*^a, for computing and visualizing the lovely results.

Thurston's Circle Packing Conjecture Theorem. *The conformal mapping of a bounded simply connected domain onto the unit disc can be approximated with a circle packing.*

In summary, we have the following:

1. An essentially unique circle packing for a spherical triangulation exists.
2. The circle packing can be computationally realized using *CirclePack*.
3. As the number of circles increase, a circle packing analogue will converge towards its continuous counterpart [4].

^a<http://www.math.utk.edu/~kens/CirclePack/>; special thanks to Ken Stephenson, University of Tennessee, for developing and maintaining this freely available software.

A DISCRETE MODEL

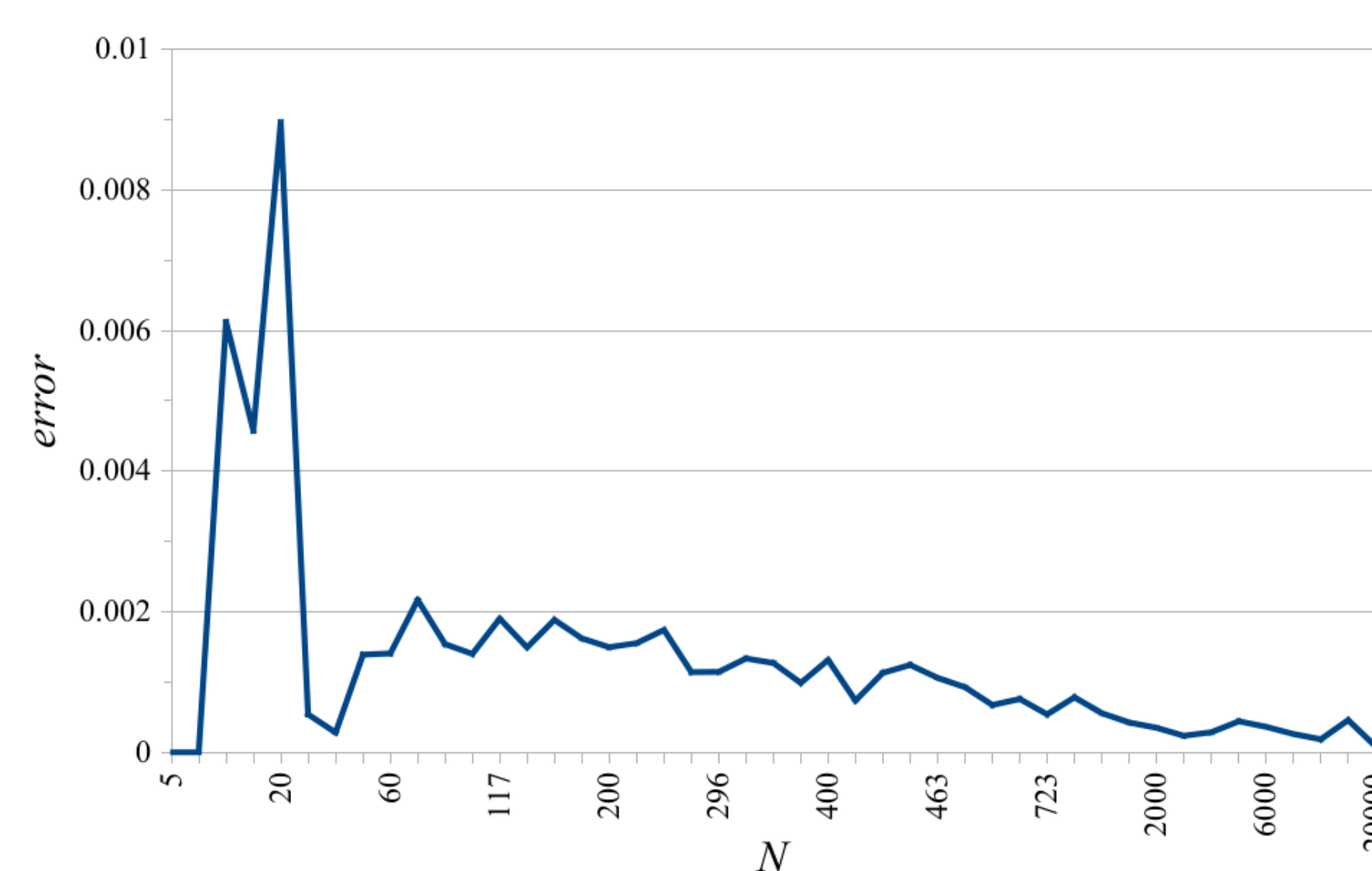
The Thomson problem is well studied, and there is a large catalog of current minimal energy distributions [5]. These point sets are stored in the interactive web applet, *thomsonapplet*^a, which can also generate random point sets, measure the energy, and implement a wide array of algorithms.

To construct a circle packing model for N energy points on the sphere we did the following:

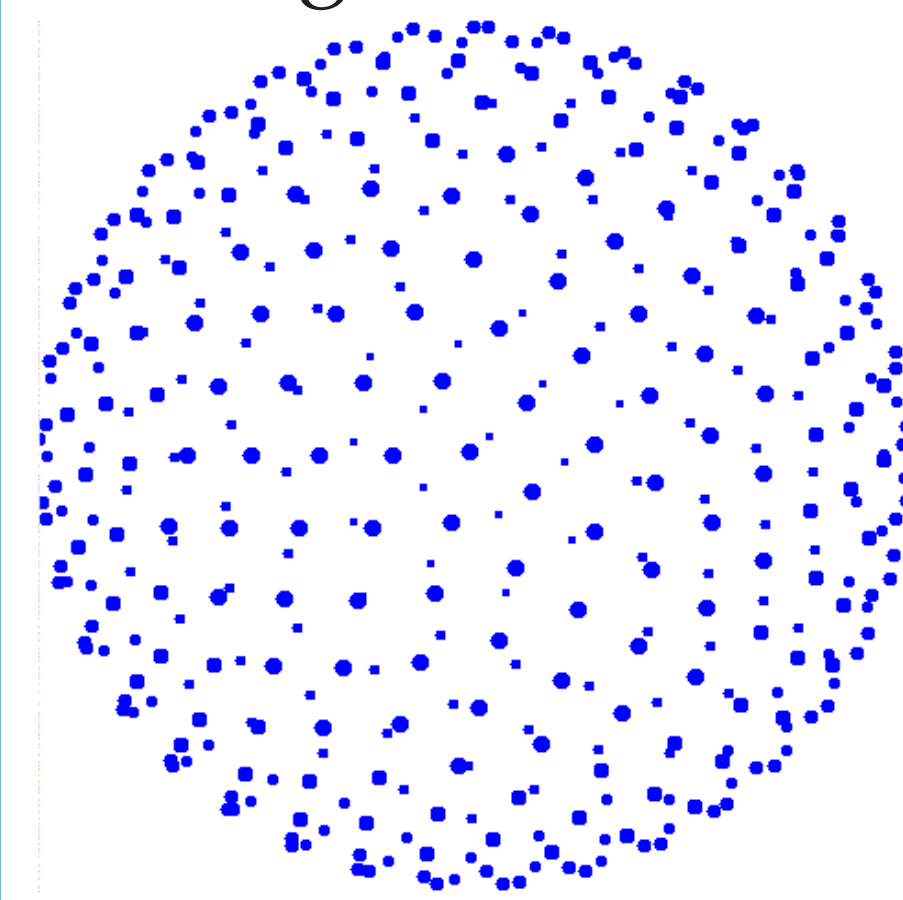
1. Find the Delaunay triangulations for N points using data from *thomsonapplet*.
2. Compute the circle packing for this triangulation using *CirclePack*.
3. Normalize the packing by applying a Möbius map such that the centroid of the circle centers in $\mathbb{R}^3$ is at the origin.

The *error* of this model was measured as,

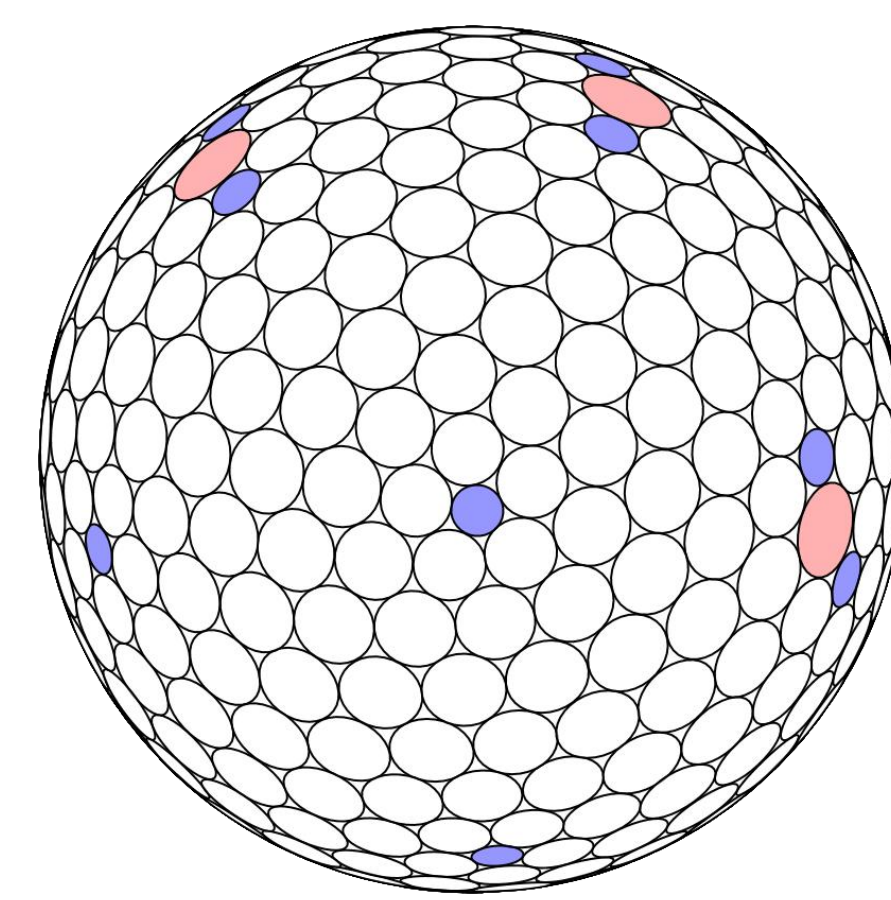
$$\text{error} = \frac{(\text{energy from circle packing}) - (\text{energy})}{\text{energy}}$$



The Figure above shows the error for minimal known N distributions. The error from our experiments never exceeded 0.0062, and as expected the error tended toward 0 for large values of N . The error introduced by the discreteness of circle packings is limited for even very small N and inconsequential for large N .



A minimal distribution



A circle packing model

^a<http://thomson.phy.syr.edu>, Cris Cecka, Primary Author. ccecka@seas.harvard.edu

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COMBINATORICS

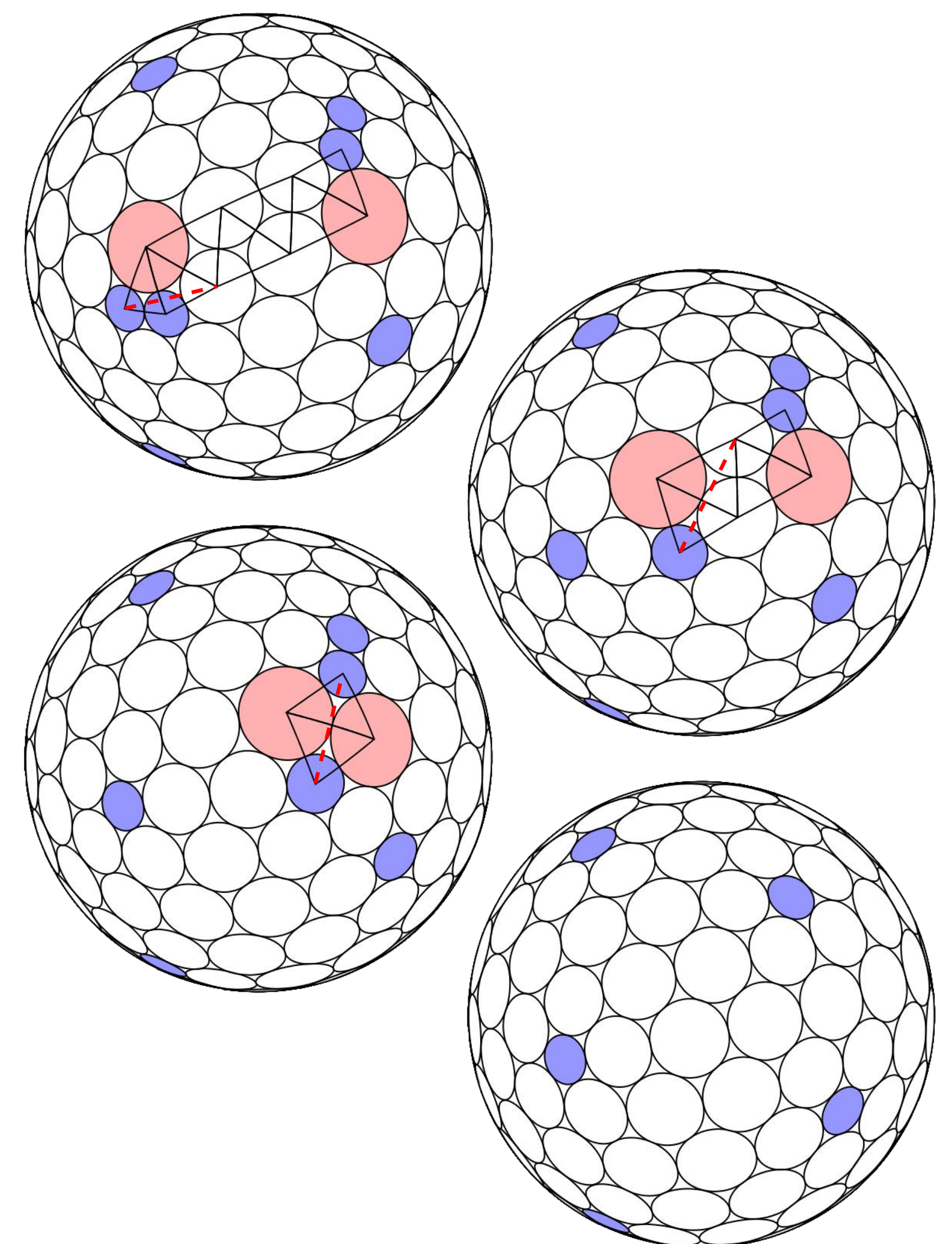
Repelling points on a flat surface naturally adopt a hexagonal pattern. However the topology of the sphere precludes a triangulation where every vertex has 6 neighbors. Try wrapping a sheet of paper around a basketball –imperfections in your wrapping job are unavoidable.

The Euler characteristic states that there will be 12 such "imperfections" where a vertex has 5 neighbors. The truncated icosahedron, a soccer ball, is a classic example. Thus a combinatorial perspective may shed new light on this problem.

MANIPULATIONS

Now it is demonstrated how manipulating the combinatorics of circle packings can be used to reduce energy. A Whitehead move, or "flip", replaces one interior edge with another in the union of two edge-sharing faces. Such a modification does not change the number of points -only the combinatorics locally. However these local changes result in a new circle packing and thus invoke a global geometric change.

A common issue with optimization algorithms is settling into a local but not a global minimum. In the figures below, two defects are separated by an area of degree six vertices. Following a sequence of flips, the defects are then combined and neutralized. Locally such defects may be difficult to fix, but the "ladder" bridges the two local imperfections.



A defect is eliminated through a series of flips.

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